

Circulatory systems link exchange surfaces with cells throughout the body

The molecular trade that an animal carries out with its environment must ultimately involve every cell in the body. Required resources, such as nutrients and oxygen (O_2), enter the cytoplasm by crossing the plasma membrane. Waste products, such as carbon dioxide, (CO_2), exit the cell by crossing the same membrane.

Small molecules in and around cells, including O_2 and CO_2 , undergo **diffusion**, which is random thermal motion (see Concept 7.3). When there is a difference in the concentration of a gas or other substance, such as between a cell and its immediate surroundings, diffusion can result in net movement.

Unicellular organisms exchange materials directly with the external environment via diffusion across the plasma membrane. For most multicellular organisms, however, direct exchange between every cell and the environment is not possible. Furthermore, net movement by diffusion is very slow for distances of more than a few millimeters. That's because the time it takes for a substance to diffuse from one place to another is proportional to the *square* of the distance. For example, a quantity of glucose that takes 1 second to diffuse $100\ \mu\text{m}$ will take 100 seconds to diffuse 1 mm and almost 3 hours to diffuse 1 cm.

Given that net movement by diffusion is rapid only over very small distances, how does each cell of an animal participate in exchange? Natural selection has resulted in two basic adaptations that permit efficient exchange for all of an animal's cells.

One adaptation for efficient exchange is a simple body plan that places many or all cells in direct contact with the environment. Each cell can thus exchange materials directly with the surrounding medium. Such an arrangement is characteristic of certain invertebrates, including cnidarians and flatworms. Animals that lack a simple body plan display an alternative adaptation for efficient exchange: a circulatory system. Such systems move fluid between each cell's immediate surroundings and the body tissues. As a result, exchange with the environment and exchange with body tissues both occur over very short distances.

In most animals, the circulatory system is functionally linked to the exchange of gases with the environment and with body cells. For this reason, we'll discuss systems for circulation and gas exchange together in this chapter. By considering examples of these systems from a range of species, we'll explore their common elements as well as their extensive variation.

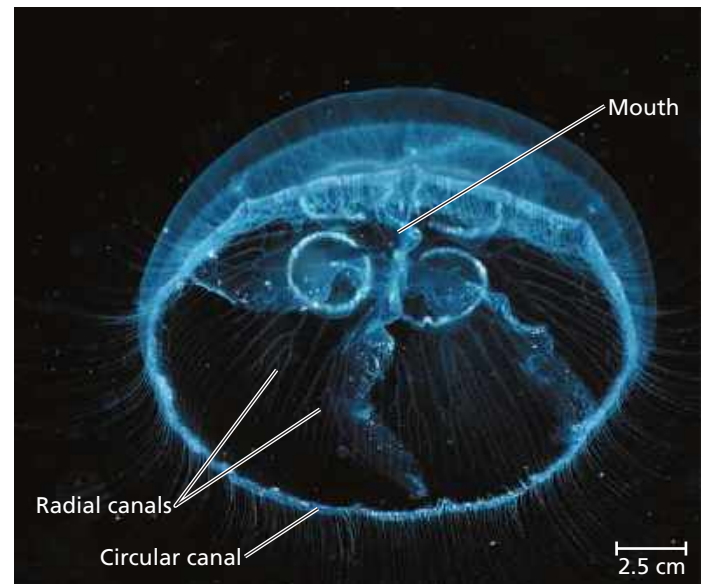
Gastrovascular Cavities

Let's begin by looking at some animals whose body shapes put many of their cells into contact with their environment,

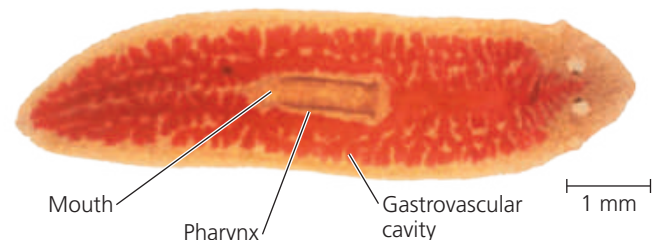
enabling them to live without a distinct circulatory system. In hydras, jellies, and other cnidarians, a central **gastrovascular cavity** functions in the distribution of substances throughout the body, as well as in digestion (see Figure 41.6). An opening at one end connects the cavity to the surrounding water. In a hydra, thin branches of the gastrovascular cavity extend into the animal's tentacles. In jellies and some other cnidarians, the gastrovascular cavity has a much more elaborate branching pattern (**Figure 42.2a**).

In animals with a gastrovascular cavity, fluid bathes both the inner and outer tissue layers, facilitating exchange of gases and cellular waste. Only the cells lining the cavity have direct access to nutrients released by digestion. However, because the body wall is a mere two cells thick, nutrients only need to diffuse a short distance to reach the cells of the outer tissue layer.

▼ **Figure 42.2** Diversity of gastrovascular cavities.



(a) The moon jelly *Aurelia*, a cnidarian. The jelly is viewed here from its underside (oral surface). The mouth leads to an elaborate gastrovascular cavity that consists of radial canals leading to and from a circular canal. Ciliated cells lining the canals circulate fluid within the cavity.



(b) The planarian *Dugesia*, a flatworm. The mouth and pharynx on the ventral side lead to the highly branched gastrovascular cavity, stained dark red in this specimen (LM).

WHAT IF? Suppose a gastrovascular cavity were open at two ends, with fluid entering one end and leaving the other. How would this affect the cavity's functions in gas exchange and digestion?